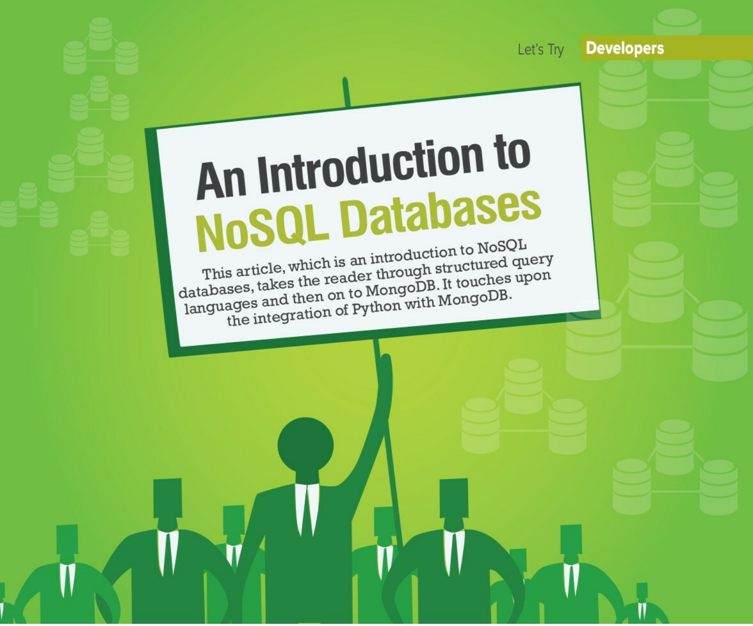

# An Introduction to NoSQL Databases

This article, which is an introduction to NoSQL databases, takes the reader through structured query languages and then on to MongoDB. It touches upon the integration of Python with MongoDB.

With the growth and development of cloud computing technology there has been a corresponding rise in NoSQL databases. In order to understand NoSQL databases, we have to study the need for them. Before one heard about NoSQL, there were many types of databases and related technologies. Most of these were relational databases, and to work on these effectively, SQL or structured query languages were developed. But these databases, as well as SQL, posed complex problems, for which developers tried to find alternate solutions. Some major problems were rigid schema designs, joins, child-parent dependencies, and so on. Developers had to look for alternatives that would ease these common problems with relational databases and SQL. Hence, 'Not only SQL' or NoSQL came into existence, which offered support for SQL-like query languages, besides other features. Some of its key features are:

- Dynamic schema design or non-schema
- No dependency (non-relational model)
- Flexibility

The following are the types of NoSQL databases (DBs) commonly found in today's market:

- Key-value databases
- Document databases
- Column family store databases
- Graph databases

In this tutorial, we will start looking into different databases, beginning with a document based DB called MongoDB, which is quite popular among developers worldwide.

## MongoDB

This is a document based NoSQL DB. It has become quite popular in recent years and many big organisations are major supporters of it. It was initially launched in 2009, and is developed and distributed by the 10gen organisation. It is open source and is freely available for developers to explore and learn.